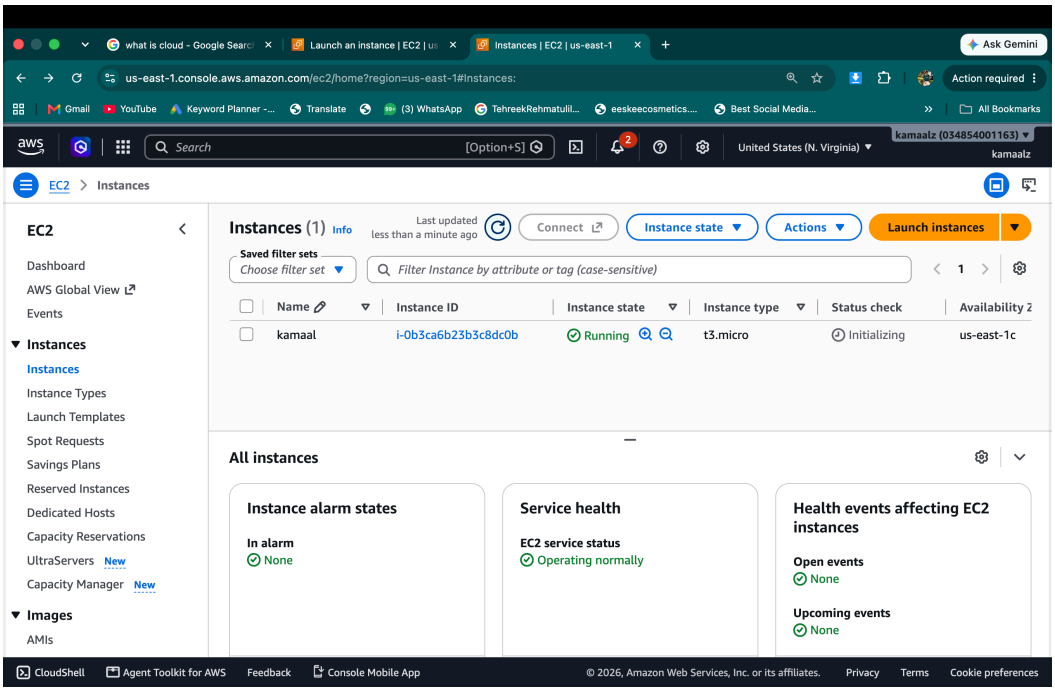
Launch one EC2 instance using Amazon Linux 2




```
sharafkamaal@192 devops-techiehorizon % ssh -i aws.pem ec2-user@44.223.68.126
The authenticity of host '44.223.68.126 (44.223.68.126)' can't be established.
ED25519 key fingerprint is: SHA256:85F5vdgbllsVg2nGeV85eSa20/i83Z3rJ6I0Q3gtd/c
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.223.68.126' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-26-145 ~]$
```

Install Docker

step1:Sudo yum install docker

```
[ec2-user@ip-172-31-26-145 ~]$ yum install docker.x86_64
Error: This command has to be run with superuser privileges (under the root user on most systems).
[ec2-user@ip-172-31-26-145 ~]$ sudo su -
[root@ip-172-31-26-145 ~]#  yum install docker.x86_64
Amazon Linux 2023 Kernel Livepatch repository                                                    806 kB/s |  89 kB     00:00
Dependencies resolved.
================================================================================================================================
 Package                          Architecture          Version                                  Repository                 Size
================================================================================================================================
Installing:
 docker                           x86_64                25.0.18-1.amzn2023.0.2                   amazonlinux                49 M
Installing dependencies:
 container-selinux                noarch                4:2.245.0-1.amzn2023                     amazonlinux                58 k
 containerd                       x86_64                2.3.7-1.amzn2023.0.1                     amazonlinux                24 M
 iptables-libs                    x86_64                1.8.8-3.amzn2023.0.2                     amazonlinux               401 k
 iptables-nft                     x86_64                1.8.8-3.amzn2023.0.2                     amazonlinux               183 k
 libcgroup                        x86_64                3.0-1.amzn2023.0.1                       amazonlinux                75 k
 libnetfilter_conntrack           x86_64                1.0.8-2.amzn2023.0.2                     amazonlinux                58 k
 libnfnetlink                     x86_64                1.0.1-19.amzn2023.0.2                    amazonlinux                30 k
 libnftnl                         x86_64                1.2.2-2.amzn2023.0.2                     amazonlinux                84 k
 pigz                             x86_64                2.5-1.amzn2023.0.4                       amazonlinux                83 k
 runc                             x86_64                1.4.3-1.amzn2023.0.3                     amazonlinux               3.7 M

Transaction Summary
================================================================================================================================
Install  11 Packages

Total download size: 77 M
Installed size: 290 M
Is this ok [y/N]: y
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm                                          1.7 MB/s |  58 kB     00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                                            9.6 MB/s | 401 kB     00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                                             4.9 MB/s | 183 kB     00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                                                  2.1 MB/s |  75 kB     00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm                                   1.9 MB/s |  58 kB     00:00
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm                                            1.1 MB/s |  30 kB     00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                                                 2.5 MB/s |  84 kB     00:00
(8/11): containerd-2.3.7-1.amzn2023.0.1.x86_64.rpm                                               68 MB/s |  24 MB     00:00
(9/11): pigz-2.5-1.amzn2023.0.4.x86_64.rpm                                                       733 kB/s |  83 kB     00:00
(10/11): runc-1.4.3-1.amzn2023.0.3.x86_64.rpm                                                    62 MB/s | 3.7 MB     00:00
```

```
Complete!
[root@ip-172-31-26-145 ~]# java -jar jenkins.war --httpPort=8080
Running from: /root/jenkins.war
webroot: /root/.jenkins/war
2026-10-28 10:52:22.996+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-10-28 10:52:23.630+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-10-28 10:52:23.731+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.13; built: 2026-09-04T06:53:08.31
1Z; git: e99386749a0a4888cd9b555ffc96a476fd48d94e; jvm 21.0.13+10-LTS
2026-10-28 10:52:24.379+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not fin
d org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-10-28 10:52:24.442+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-10-28 10:52:25.113+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: /root/.jenkins found at
: $user.home/.jenkins
2026-10-28 10:52:25.257+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHand
ler@7ae9a33a{Jenkins v2.580.1,/,b=file:///root/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandl
er@301aa982{STARTED}}
2026-10-28 10:52:25.285+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@1b7cae6f{HTTP/1.
1, (http/1.1)}{0.0.0.0:8080}
2026-10-28 10:52:25.285+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@21f8e55f{STARTING}[12.1.
13,sto=0] @1558ms
2026-10-28 10:52:25.287+0000 [id=25]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-10-28 10:52:25.482+0000 [id=24]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.580.1
2026-10-28 10:52:25.603+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-10-28 10:52:25.610+0000 [id=31]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-10-28 10:52:26.751+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-10-28 10:52:26.759+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-10-28 10:52:26.765+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-10-28 10:52:27.008+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-10-28 10:52:27.009+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-10-28 10:52:27.009+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-10-28 10:52:27.011+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-10-28 10:52:27.087+0000 [id=48]    INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-10-28 10:52:27.631+0000 [id=31]    INFO    jenkins.install.SetupWizard#init:
[LF]>
[LF]> ************************************************************
[LF]> ************************************************************
[LF]> ************************************************************
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
```

Install Jenkins

Step1:
Add the Jenkins repo using the following command:
sudo wget -O /etc/yum.repos.d/jenkins.repo \
sudo wget -O /etc/yum.repos.d/jenkins.repo \
Step2: Import a key file from Jenkins-CI to enable installation from the package:
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io.key

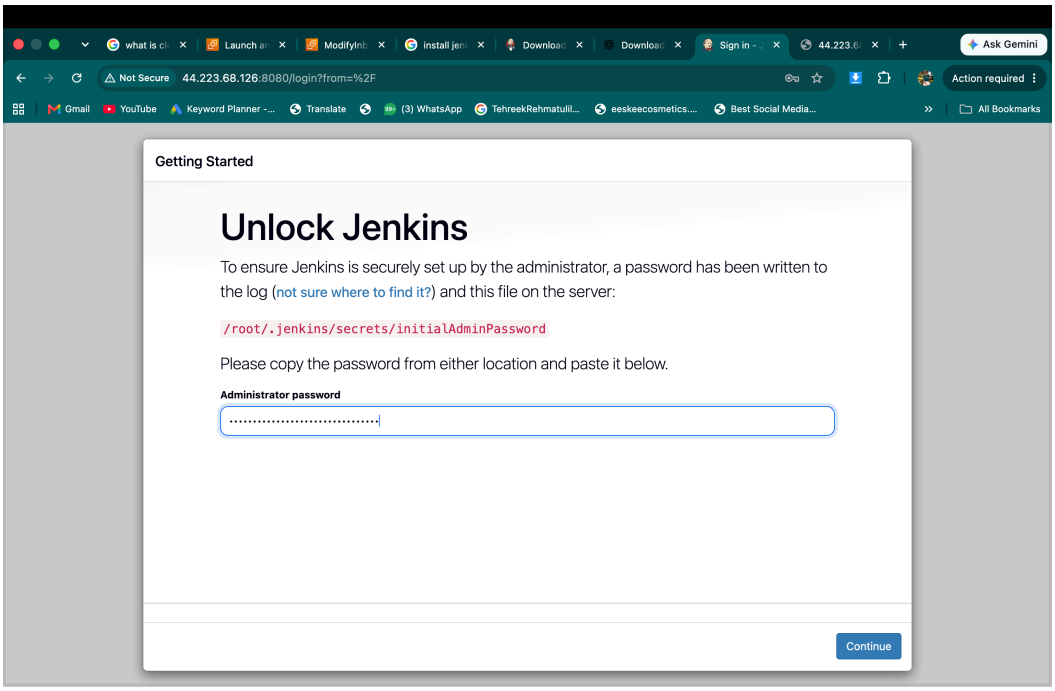
```
  Installing       : runc-1.4.3-1.amzn2023.0.3.x86_64                                                                   1/11
  Installing       : containerd-2.3.7-1.amzn2023.0.1.x86_64                                                             2/11
  Running scriptlet: containerd-2.3.7-1.amzn2023.0.1.x86_64                                                             2/11
  Installing       : pigz-2.5-1.amzn2023.0.4.x86_64                                                                     3/11
  Installing       : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                                               4/11
  Installing       : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64                                                          5/11
  Installing       : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64                                                 6/11
  Installing       : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                                                          7/11
  Installing       : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                                                           8/11
  Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                                                           8/11
  Installing       : libcgroup-3.0-1.amzn2023.0.1.x86_64                                                                9/11
  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                                                      10/11
  Installing       : container-selinux-4:2.245.0-1.amzn2023.noarch                                                      10/11
  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                                                      10/11
  Running scriptlet: docker-25.0.18-1.amzn2023.0.2.x86_64                                                               11/11
  Installing       : docker-25.0.18-1.amzn2023.0.2.x86_64                                                               11/11
  Running scriptlet: docker-25.0.18-1.amzn2023.0.2.x86_64                                                               11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

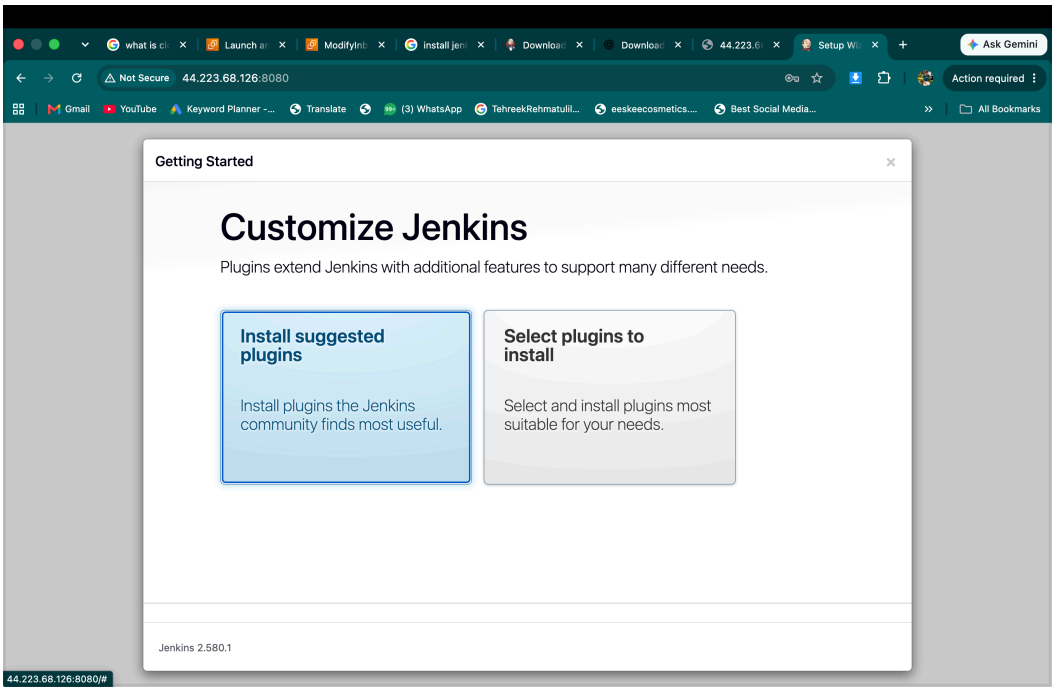
  Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch                                                      11/11
  Running scriptlet: docker-25.0.18-1.amzn2023.0.2.x86_64                                                               11/11
  Verifying        : container-selinux-4:2.245.0-1.amzn2023.noarch                                                       1/11
  Verifying        : containerd-2.3.7-1.amzn2023.0.1.x86_64                                                              2/11
  Verifying        : docker-25.0.18-1.amzn2023.0.2.x86_64                                                                3/11
  Verifying        : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64                                                           4/11
  Verifying        : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64                                                            5/11
  Verifying        : libcgroup-3.0-1.amzn2023.0.1.x86_64                                                                 6/11
  Verifying        : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64                                                  7/11
  Verifying        : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64                                                           8/11
  Verifying        : libnftnl-1.2.2-2.amzn2023.0.2.x86_64                                                                9/11
  Verifying        : pigz-2.5-1.amzn2023.0.4.x86_64                                                                     10/11
  Verifying        : runc-1.4.3-1.amzn2023.0.3.x86_64                                                                   11/11

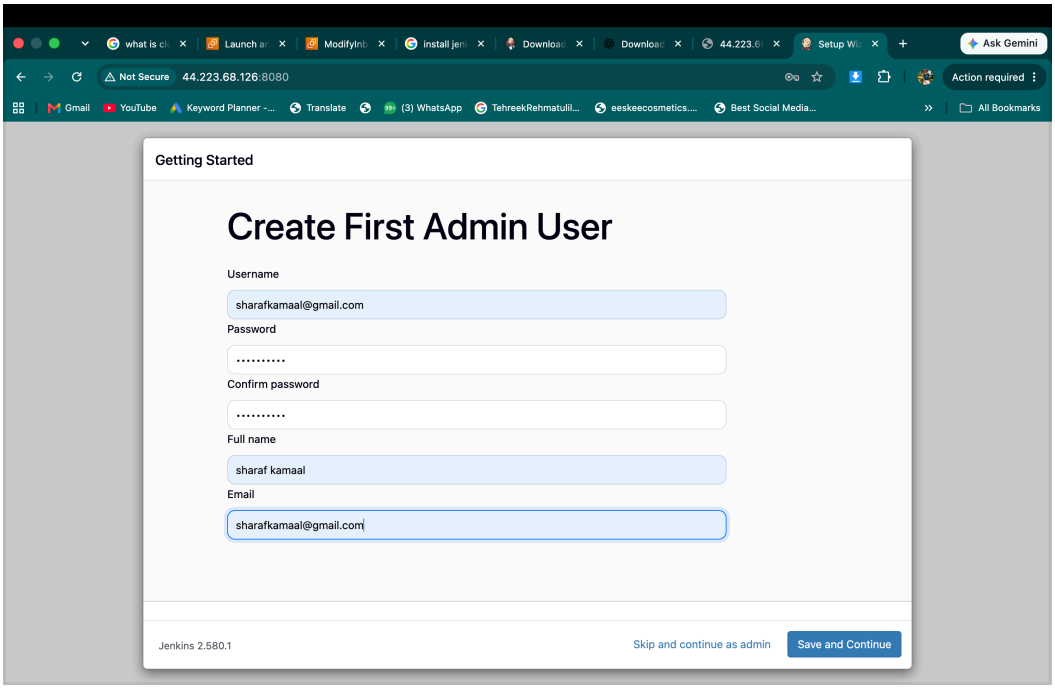
Installed:
  container-selinux-4:2.245.0-1.amzn2023.noarch      containerd-2.3.7-1.amzn2023.0.1.x86_64    docker-25.0.18-1.amzn2023.0.2.x86_64
  iptables-libs-1.8.8-3.amzn2023.0.2.x86_64          iptables-nft-1.8.8-3.amzn2023.0.2.x86_64  libcgroup-3.0-1.amzn2023.0.1.x86_64
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64 libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  pigz-2.5-1.amzn2023.0.4.x86_64                     runc-1.4.3-1.amzn2023.0.3.x86_64

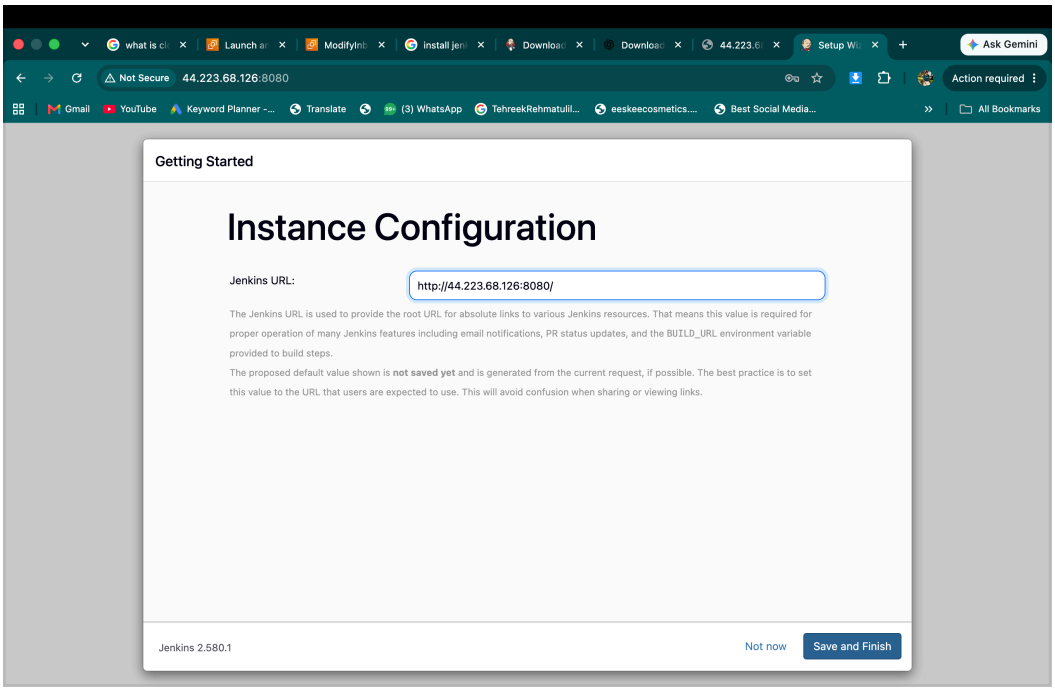
Complete!
[root@ip-172-31-26-145 ~]#
```

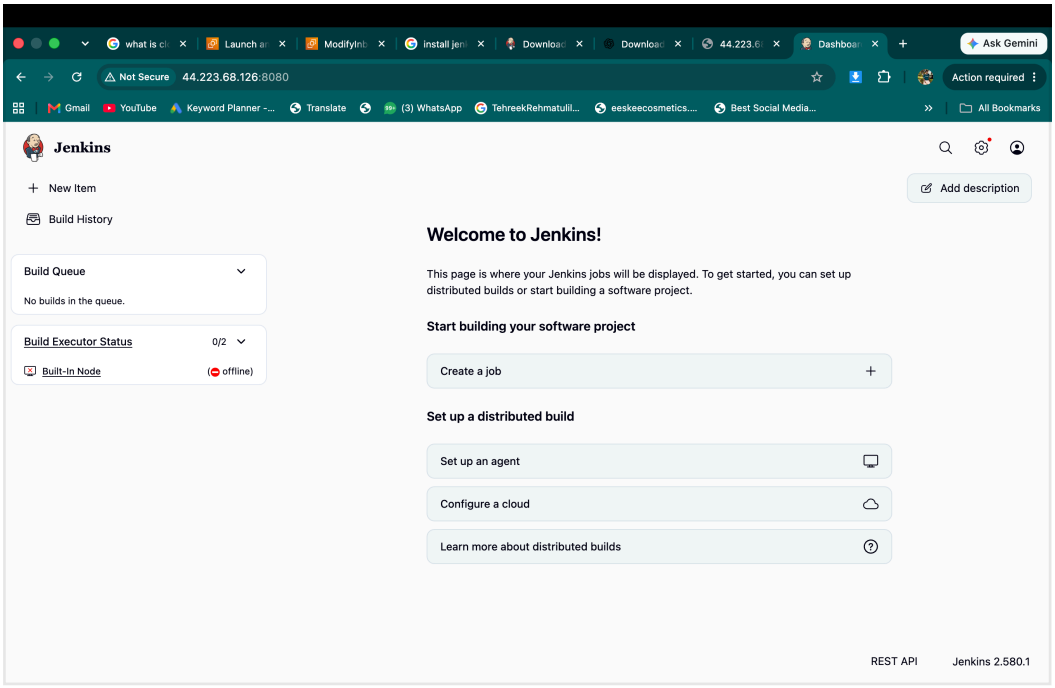
```
Complete!
[root@ip-172-31-26-145 ~]# java -jar jenkins.war --httpPort=8080
Running from: /root/jenkins.war
webroot: /root/.jenkins/war
2026-10-28 10:52:22.996+0000 [id=1]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-10-28 10:52:23.630+0000 [id=1]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-10-28 10:52:23.731+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.13; built: 2026-09-04T06:53:08.311Z; git: e99386749a0a4888cd9b555ffc96a476fd48d94e; jvm 21.0.13+10-LTS
2026-10-28 10:52:24.379+0000 [id=1]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-10-28 10:52:24.442+0000 [id=1]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-10-28 10:52:25.113+0000 [id=1]     INFO    hudson.WebAppMain#contextInitialized: Jenkins home directory: /root/.jenkins found at: $user.home/.jenkins
2026-10-28 10:52:25.257+0000 [id=1]     INFO    o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@7ae9a33a{Jenkins v2.580.1,/,b=file:///root/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@301aa982{STARTED}}
2026-10-28 10:52:25.285+0000 [id=1]     INFO    o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@1b7cae6f{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-10-28 10:52:25.285+0000 [id=1]     INFO    org.eclipse.jetty.server.Server#doStart: Started oejs.Server@21f8e55f{STARTING}[12.1.13,sto=0] @1558ms
2026-10-28 10:52:25.287+0000 [id=25]    INFO    winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-10-28 10:52:25.482+0000 [id=24]    INFO    jenkins.model.Jenkins#<init>: Starting version 2.580.1
2026-10-28 10:52:25.603+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-10-28 10:52:25.610+0000 [id=31]    INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-10-28 10:52:26.751+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-10-28 10:52:26.759+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-10-28 10:52:26.765+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-10-28 10:52:27.008+0000 [id=34]    INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-10-28 10:52:27.009+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-10-28 10:52:27.009+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-10-28 10:52:27.011+0000 [id=33]    INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-10-28 10:52:27.087+0000 [id=48]    INFO    hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-10-28 10:52:27.631+0000 [id=31]    INFO    jenkins.install.SetupWizard#init:
[LF]>
[LF]> ************************************************************
[LF]> ************************************************************
[LF]> ************************************************************
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
```


Not Secure 44.223.68.126:8080/login?from=%2F
Action required
Getting Started
Unlock Jenkins
To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:
/root/.jenkins/secrets/initialAdminPassword
Please copy the password from either location and paste it below.
Administrator password
Continue


Not Secure 44.223.68.126:8080
Action required
Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.580.1
44.223.68.126:8080/#


Not Secure 44.223.68.126:8080
Action required
Getting Started
Create First Admin User
Username
sharafkamaal@gmail.com
Password
Confirm password
Full name
sharaf kamaal
Email
sharafkamaal@gmail.com
Jenkins 2.580.1
Skip and continue as admin
Save and Continue


Not Secure 44.223.68.126:8080
Action required
Getting Started
Instance Configuration
Jenkins URL:
http://44.223.68.126:8080/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.580.1
Not now
Save and Finish



Install Apache

Step1: yum install httpd

```
[root@ip-172-31-26-145 ~]# yum install httpd
Last metadata expiration check: 0:13:45 ago on Wed Oct 28 10:46:14 2026.
Dependencies resolved.
================================================================================================================================================
 Package                          Architecture          Version                                  Repository                  Size
================================================================================================================================================
Installing:
 httpd                            x86_64                2.4.69-1.amzn2023.0.2                    amazonlinux                 45 k
Installing dependencies:
 apr                              x86_64                1.7.5-1.amzn2023.0.4                     amazonlinux                129 k
 apr-util                         x86_64                1.6.5-1.amzn2023.0.1                     amazonlinux                100 k
 apr-util-lmdb                    x86_64                1.6.5-1.amzn2023.0.1                     amazonlinux                 15 k
 generic-logos-httpd              noarch                18.0.0-12.amzn2023.0.3                   amazonlinux                 19 k
 httpd-core                       x86_64                2.4.69-1.amzn2023.0.2                    amazonlinux                1.4 M
 httpd-filesystem                 noarch                2.4.69-1.amzn2023.0.2                    amazonlinux                 12 k
 httpd-tools                      x86_64                2.4.69-1.amzn2023.0.2                    amazonlinux                 79 k
 mailcap                          noarch                2.1.49-3.amzn2023.0.3                    amazonlinux                 33 k
Installing weak dependencies:
 apr-util-openssl                 x86_64                1.6.5-1.amzn2023.0.1                     amazonlinux                 17 k
 mod_http2                        x86_64                1:2.4.69-1.amzn2023.0.2                  amazonlinux                130 k
 mod_lua                          x86_64                2.4.69-1.amzn2023.0.2                    amazonlinux                 58 k

Transaction Summary
================================================================================================================================================
Install  12 Packages

Total download size: 2.0 M
Installed size: 6.1 M
Is this ok [y/N]: y
Downloading Packages:
(1/12): apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64.rpm                                437 kB/s |  15 kB     00:00
(2/12): apr-util-1.6.5-1.amzn2023.0.1.x86_64.rpm                                     2.5 MB/s | 100 kB     00:00
(3/12): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                                          3.0 MB/s | 129 kB     00:00
(4/12): apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64.rpm                             640 kB/s |  17 kB     00:00
(5/12): httpd-2.4.69-1.amzn2023.0.2.x86_64.rpm                                       1.8 MB/s |  45 kB     00:00
(6/12): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                        593 kB/s |  19 kB     00:00
(7/12): httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch.rpm                            309 kB/s |  12 kB     00:00
(8/12): httpd-tools-2.4.69-1.amzn2023.0.2.x86_64.rpm                                 2.0 MB/s |  79 kB     00:00
(9/12): httpd-core-2.4.69-1.amzn2023.0.2.x86_64.rpm                                   18 MB/s | 1.4 MB     00:00
(10/12): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm                                    868 kB/s |  33 kB     00:00
(11/12): mod_http2-2.4.69-1.amzn2023.0.2.x86_64.rpm                                  3.2 MB/s | 130 kB     00:00
```

```
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                    1/1
  Installing       : apr-1.7.5-1.amzn2023.0.4.x86_64                                                                    1/12
  Installing       : apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64                                                          2/12
  Installing       : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64                                                       3/12
  Installing       : apr-util-1.6.5-1.amzn2023.0.1.x86_64                                                               4/12
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                               5/12
  Running scriptlet: httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch                                                      6/12
  Installing       : httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch                                                      6/12
  Installing       : httpd-tools-2.4.69-1.amzn2023.0.2.x86_64                                                           7/12
  Installing       : httpd-core-2.4.69-1.amzn2023.0.2.x86_64                                                            8/12
  Installing       : mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64                                                           9/12
  Installing       : mod_lua-2.4.69-1.amzn2023.0.2.x86_64                                                              10/12
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                 11/12
  Installing       : httpd-2.4.69-1.amzn2023.0.2.x86_64                                                                12/12
  Running scriptlet: httpd-2.4.69-1.amzn2023.0.2.x86_64                                                                12/12
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                                                    1/12
  Verifying        : apr-util-1.6.5-1.amzn2023.0.1.x86_64                                                               2/12
  Verifying        : apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64                                                          3/12
  Verifying        : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64                                                       4/12
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                  5/12
  Verifying        : httpd-2.4.69-1.amzn2023.0.2.x86_64                                                                 6/12
  Verifying        : httpd-core-2.4.69-1.amzn2023.0.2.x86_64                                                            7/12
  Verifying        : httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch                                                      8/12
  Verifying        : httpd-tools-2.4.69-1.amzn2023.0.2.x86_64                                                           9/12
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                              10/12
  Verifying        : mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64                                                          11/12
  Verifying        : mod_lua-2.4.69-1.amzn2023.0.2.x86_64                                                              12/12

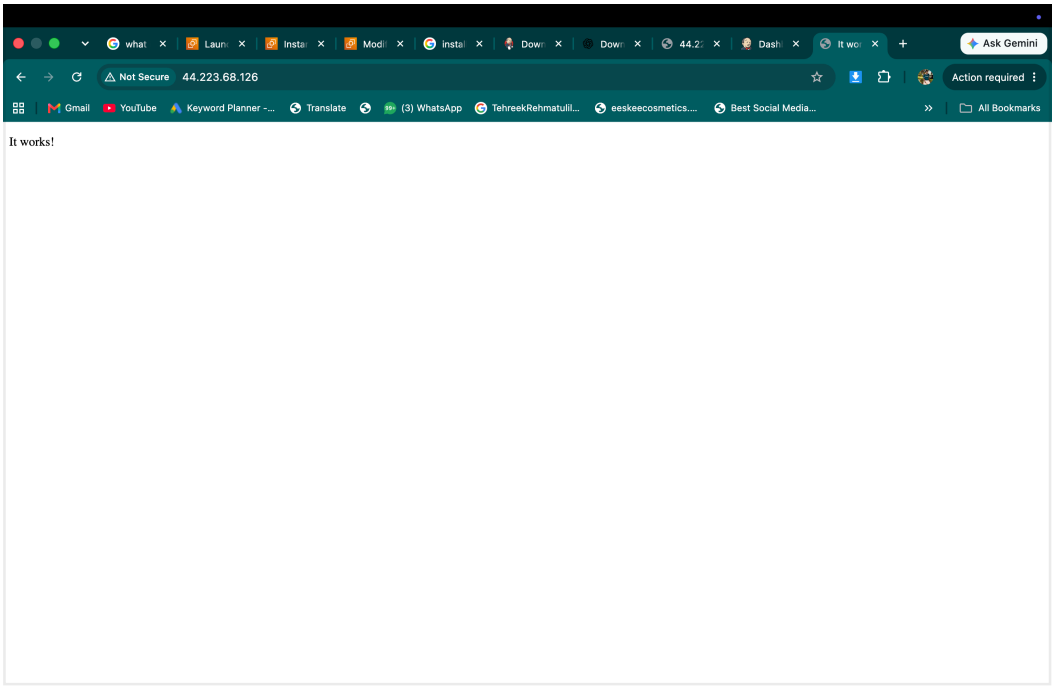
Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                          apr-util-1.6.5-1.amzn2023.0.1.x86_64
  apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64                apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch        httpd-2.4.69-1.amzn2023.0.2.x86_64
  httpd-core-2.4.69-1.amzn2023.0.2.x86_64                  httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch
  httpd-tools-2.4.69-1.amzn2023.0.2.x86_64                 mailcap-2.1.49-3.amzn2023.0.3.noarch
  mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64                 mod_lua-2.4.69-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-26-145 ~]#
```

step 2: systemctl start httpd

```
[root@ip-172-31-26-145 ~]# systemctl start http
Failed to start http.service: Unit http.service not found.
[root@ip-172-31-26-145 ~]# systemctl start httpd
[root@ip-172-31-26-145 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-28 11:01:09 UTC; 10s ago
       Docs: man:httpd.service(8)
   Main PID: 29144 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 14.1M
        CPU: 67ms
     CGroup: /system.slice/httpd.service
             ├─29144 /usr/sbin/httpd -DFOREGROUND
             ├─29145 /usr/sbin/httpd -DFOREGROUND
             ├─29146 /usr/sbin/httpd -DFOREGROUND
             ├─29153 /usr/sbin/httpd -DFOREGROUND
             └─29168 /usr/sbin/httpd -DFOREGROUND

Oct 28 11:01:09 ip-172-31-26-145.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 28 11:01:09 ip-172-31-26-145.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 28 11:01:09 ip-172-31-26-145.ec2.internal httpd[29144]: Server configured, listening on: port 80
[root@ip-172-31-26-145 ~]#
```

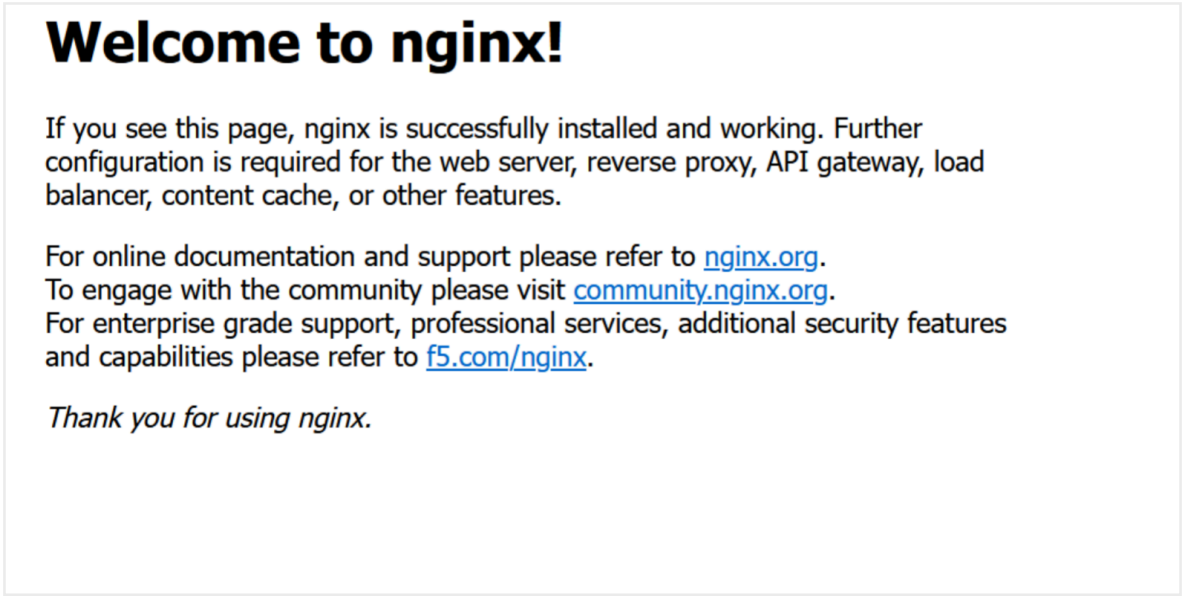


Install Nginx
Step1: sudo yum install nginx

```
[root@ip-172-31-40-38 ec2-user]# yum install nginx
Last metadata expiration check: 0:24:25 ago on Tue Oct 27 10:11:18 2026.
Dependencies resolved.
================================================================================
 Package                    Architecture                Version
================================================================================
Installing:
 nginx                      x86_64                      1:1.30.6-1.amzn2023.0.1
Installing dependencies:
 gperftools-libs            x86_64                      2.9.1-1.amzn2023.0.3
 libunwind                  x86_64                      1.4.0-5.amzn2023.0.3
 nginx-core                 x86_64                      1:1.30.6-1.amzn2023.0.1
 nginx-filesystem           noarch                      1:1.30.6-1.amzn2023.0.1
 nginx-mimetypes            noarch                      2.1.49-3.amzn2023.0.3

Transaction Summary
================================================================================
Install  6 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Is this ok [y/N]: y
Downloading Packages:
(1/6): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm
(2/6): nginx-1.30.6-1.amzn2023.0.1.x86_64.rpm
(3/6): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm
(4/6): nginx-core-1.30.6-1.amzn2023.0.1.x86_64.rpm
(5/6): nginx-filesystem-1.30.6-1.amzn2023.0.1.noarch.rpm
(6/6): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm
```



Install Apache Tomcat.

Step1: sudo yum install tomcat10

```
[root@ip-172-31-26-145 ~]# sudo yum install tomcat10
Last metadata expiration check: 0:20:41 ago on Wed Oct 28 10:46:14 2026.
Dependencies resolved.
================================================================================================================================================
 Package                                  Architecture      Version                                    Repository                Size
================================================================================================================================================
Installing:
 tomcat10                                 noarch            1:10.1.60-1.amzn2023.0.1                   amazonlinux               94 k
Installing dependencies:
 ecj                                      noarch            1:4.23-1.amzn2023.0.4                      amazonlinux              2.4 M
 java-17-amazon-corretto-headless         x86_64            1:17.0.21.0.10-1.amzn2023.1                amazonlinux               91 M
 javapackages-tools                       noarch            6.0.0-7.amzn2023.0.6                       amazonlinux               25 k
 tomcat10-el-5.0-api                      noarch            1:10.1.60-1.amzn2023.0.1                   amazonlinux              109 k
 tomcat10-jsp-3.1-api                     noarch            1:10.1.60-1.amzn2023.0.1                   amazonlinux               77 k
 tomcat10-lib                             noarch            1:10.1.60-1.amzn2023.0.1                   amazonlinux              7.1 M
 tomcat10-servlet-6.0-api                 noarch            1:10.1.60-1.amzn2023.0.1                   amazonlinux              361 k
Installing weak dependencies:
 tomcat-native                            x86_64            2.0.16-1.amzn2023.0.1                      amazonlinux               45 k

Transaction Summary
================================================================================================================================================
Install  9 Packages

Total download size: 102 M
Installed size: 248 M
Is this ok [y/N]: y
Downloading Packages:
(1/9): javapackages-tools-6.0.0-7.amzn2023.0.6.noarch.rpm                                      946 kB/s |  25 kB     00:00
(2/9): ecj-4.23-1.amzn2023.0.4.noarch.rpm                                                       46 MB/s | 2.4 MB     00:00
(3/9): tomcat-native-2.0.16-1.amzn2023.0.1.x86_64.rpm                                          1.5 MB/s |  45 kB     00:00
(4/9): tomcat10-10.1.60-1.amzn2023.0.1.noarch.rpm                                              3.3 MB/s |  94 kB     00:00
(5/9): tomcat10-el-5.0-api-10.1.60-1.amzn2023.0.1.noarch.rpm                                   3.2 MB/s | 109 kB     00:00
(6/9): tomcat10-jsp-3.1-api-10.1.60-1.amzn2023.0.1.noarch.rpm                                  1.9 MB/s |  77 kB     00:00
(7/9): tomcat10-servlet-6.0-api-10.1.60-1.amzn2023.0.1.noarch.rpm                              5.7 MB/s | 361 kB     00:00
(8/9): tomcat10-lib-10.1.60-1.amzn2023.0.1.noarch.rpm                                           54 MB/s | 7.1 MB     00:00
(9/9): java-17-amazon-corretto-headless-17.0.21.0.10-1.amzn2023.1.x86_64.rpm                    70 MB/s |  91 MB     00:01
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                          74 MB/s | 102 MB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
```

```
Total                                                                                                74 MB/s | 102 MB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                          1/1
  Installing       : java-17-amazon-corretto-headless-1:17.0.21.0.10-1.amzn2023.1.x86_64                                      1/9
  Running scriptlet: java-17-amazon-corretto-headless-1:17.0.21.0.10-1.amzn2023.1.x86_64                                      1/9
  Installing       : tomcat10-el-5.0-api-1:10.1.60-1.amzn2023.0.1.noarch                                                      2/9
  Running scriptlet: tomcat10-el-5.0-api-1:10.1.60-1.amzn2023.0.1.noarch                                                      2/9
  Installing       : tomcat10-servlet-6.0-api-1:10.1.60-1.amzn2023.0.1.noarch                                                 3/9
  Running scriptlet: tomcat10-servlet-6.0-api-1:10.1.60-1.amzn2023.0.1.noarch                                                 3/9
  Installing       : tomcat10-jsp-3.1-api-1:10.1.60-1.amzn2023.0.1.noarch                                                     4/9
  Running scriptlet: tomcat10-jsp-3.1-api-1:10.1.60-1.amzn2023.0.1.noarch                                                     4/9
  Installing       : ecj-1:4.23-1.amzn2023.0.4.noarch                                                                         5/9
  Installing       : tomcat10-lib-1:10.1.60-1.amzn2023.0.1.noarch                                                             6/9
  Installing       : javapackages-tools-6.0.0-7.amzn2023.0.6.noarch                                                           7/9
  Installing       : tomcat-native-2.0.16-1.amzn2023.0.1.x86_64                                                               8/9
  Running scriptlet: tomcat10-1:10.1.60-1.amzn2023.0.1.noarch                                                                 9/9
  Installing       : tomcat10-1:10.1.60-1.amzn2023.0.1.noarch                                                                 9/9
  Running scriptlet: tomcat10-1:10.1.60-1.amzn2023.0.1.noarch                                                                 9/9
  Verifying        : ecj-1:4.23-1.amzn2023.0.4.noarch                                                                         1/9
  Verifying        : java-17-amazon-corretto-headless-1:17.0.21.0.10-1.amzn2023.1.x86_64                                      2/9
  Verifying        : javapackages-tools-6.0.0-7.amzn2023.0.6.noarch                                                           3/9
  Verifying        : tomcat-native-2.0.16-1.amzn2023.0.1.x86_64                                                               4/9
  Verifying        : tomcat10-1:10.1.60-1.amzn2023.0.1.noarch                                                                 5/9
  Verifying        : tomcat10-el-5.0-api-1:10.1.60-1.amzn2023.0.1.noarch                                                      6/9
  Verifying        : tomcat10-jsp-3.1-api-1:10.1.60-1.amzn2023.0.1.noarch                                                     7/9
  Verifying        : tomcat10-lib-1:10.1.60-1.amzn2023.0.1.noarch                                                             8/9
  Verifying        : tomcat10-servlet-6.0-api-1:10.1.60-1.amzn2023.0.1.noarch                                                 9/9

Installed:
  ecj-1:4.23-1.amzn2023.0.4.noarch                          java-17-amazon-corretto-headless-1:17.0.21.0.10-1.amzn2023.1.x86_64
  javapackages-tools-6.0.0-7.amzn2023.0.6.noarch            tomcat-native-2.0.16-1.amzn2023.0.1.x86_64
  tomcat10-1:10.1.60-1.amzn2023.0.1.noarch                  tomcat10-el-5.0-api-1:10.1.60-1.amzn2023.0.1.noarch
  tomcat10-jsp-3.1-api-1:10.1.60-1.amzn2023.0.1.noarch      tomcat10-lib-1:10.1.60-1.amzn2023.0.1.noarch
  tomcat10-servlet-6.0-api-1:10.1.60-1.amzn2023.0.1.noarch

Complete!
[root@ip-172-31-26-145 ~]#
```

Step2: after configuring port
Sudo systemctl restart tomcat10
Sudo systemctl status tomcat10

```
Complete!
[root@ip-172-31-26-145 ~]# Sudo systemctl restart tomcat10
-bash: Sudo: command not found
[root@ip-172-31-26-145 ~]# sudo systemctl restart tomcat10
[root@ip-172-31-26-145 ~]# sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-28 11:08:16 UTC; 19s ago
   Main PID: 29955 (java)
      Tasks: 29 (limit: 1059)
     Memory: 87.5M
        CPU: 2.404s
     CGroup: /system.slice/tomcat10.service
             └─29955 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.Factory=org.apache.commons.dbcp.BasicDataSourceFactory --add-o>

Oct 28 11:08:17 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:17.621 INFO [main] org.apache.catalina.startup.Versio>
Oct 28 11:08:17 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:17.621 INFO [main] org.apache.catalina.startup.Versio>
Oct 28 11:08:17 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:17.632 INFO [main] org.apache.catalina.core.AprLifecy>
Oct 28 11:08:17 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:17.640 INFO [main] org.apache.catalina.core.AprLifecy>
Oct 28 11:08:18 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:18.257 INFO [main] org.apache.coyote.AbstractProtocol>
Oct 28 11:08:18 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:18.313 INFO [main] org.apache.catalina.startup.Catali>
Oct 28 11:08:18 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:18.391 INFO [main] org.apache.catalina.core.StandardS>
Oct 28 11:08:18 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:18.391 INFO [main] org.apache.catalina.core.StandardE>
Oct 28 11:08:18 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:18.411 INFO [main] org.apache.coyote.AbstractProtocol>
Oct 28 11:08:18 ip-172-31-26-145.ec2.internal server[29955]: 28-Oct-2026 11:08:18.435 INFO [main] org.apache.catalina.startup.Catali>
lines 1-20/20 (END)
```